



Systems Design

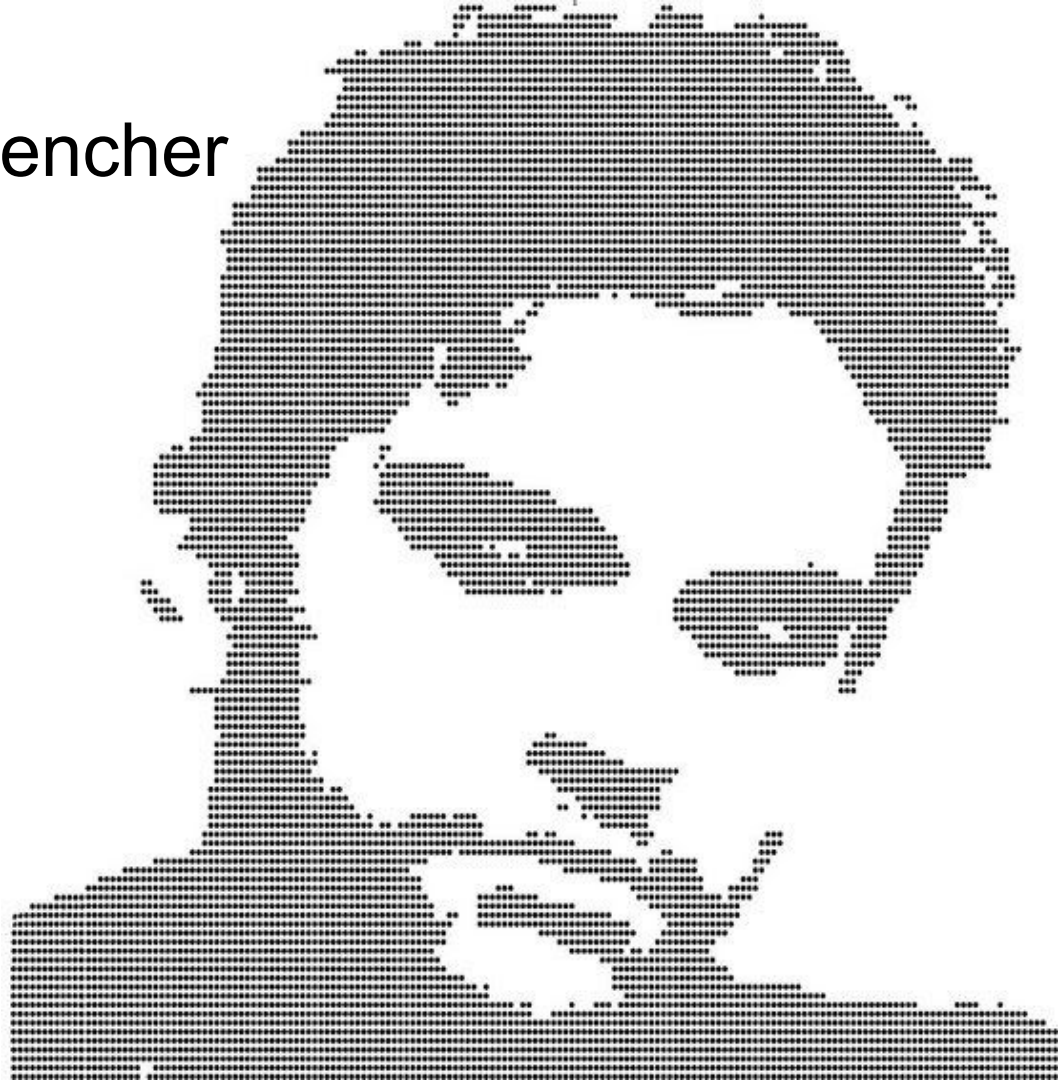
Feb 17, 2026

Pia Zaragoza

Agenda

- 6:30pm - 7:30pm: Guest Lecture w/ Ariele Faber
- 7:30pm - 7:40pm: Stretch Break
- 7:40pm - 8:00pm: Lecture
- 8:00pm - 9:20pm: Review Homework

John Provencher



Actor Maps

Why Actor Maps?

Involvement

Start to recognize who will be affected by a change process and thus who needs to be involved

Communications

To understand the mental models and values of actors so that we can better speak their language

Opportunities

Find opportunities for alliances, collaboration and recognize potential points of conflict



Explore Perspectives

Build up a better understanding of the system by looking at it from the different perspectives of the actors

Identify Gaps

Identify where gaps to the flow of information or resources are as potential intervention points

Adoption

Understand how an adoption process may take place; who would be likely to resist or promote change

Instructions

1) ***Prepare***

- Identify the topic and set clear boundaries
- Frame the system
- Identify an initial set of key actors

2) ***Facilitate a live actor mapping session***

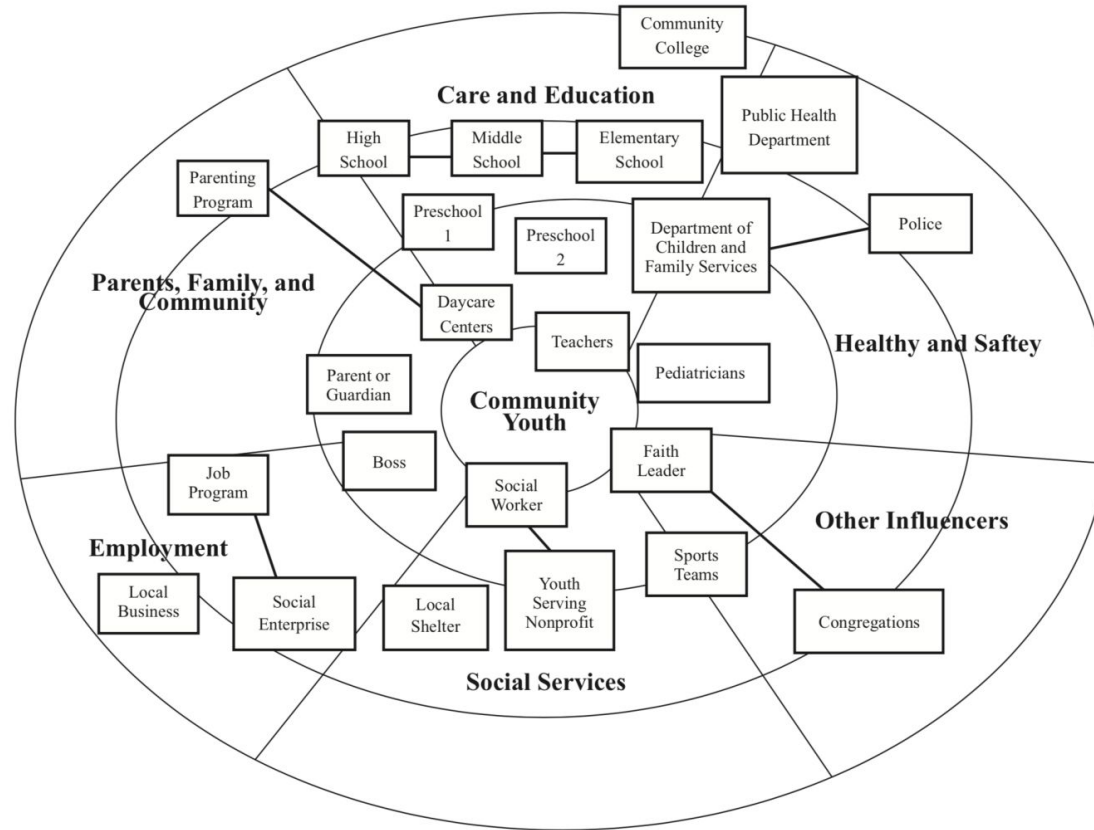
- Populate the map
- Refine the map by including additional actors
- Map engagement, relationships, and connections
- Identify momentum, blockages, and opportunities
- Discuss implications

3) ***Refine and revise the actor maps***

- Don't forget to include a key.



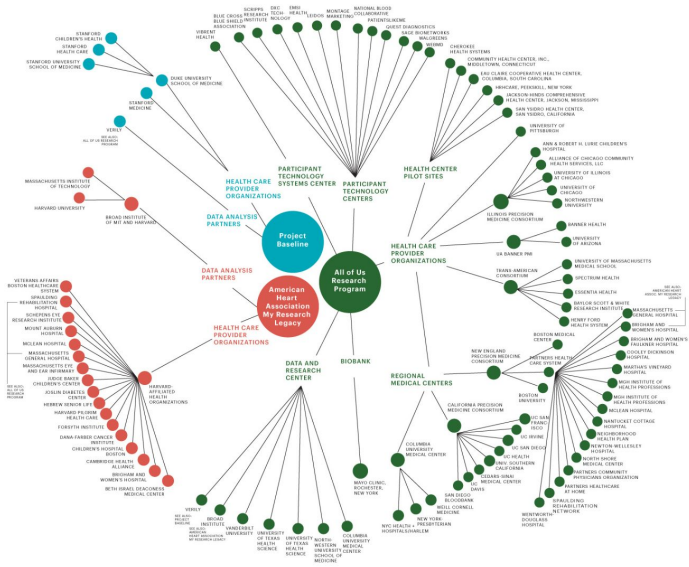
Improving the Lives of Community Youth



Precision Medicine National Actor Map

The Precision Medicine National Actor Map provides an overview of three major national precision medicine projects and the network of institutions connected to them as grantees and sub-grantees. The projects include: the **All of Us Research Program**, **My Research Legacy**, and **Project Baseline**.¹

Kadija Ferryman
Mikaela Pitcan



All of Us Research Program

The All of Us (AoU) research project aims to build a cohort of 1 million people across the United States. Participants will be able to join the study through a health care provider or directly through the program's web site. Approximately 28 large academic medical centers and Federally qualified health centers have been awarded funding to recruit and enroll a share of the 1 million participants. The project's goal is to collect data on health through physical measurements, surveys, electronic health records, and—for a subset of participants—activity trackers. The AoU is a prospective observational study. Thus, the study is not focused on assessing the efficacy of one treatment or another. AoU was initially divided into four areas: 1) health care provider organizations, 2) the Participant Technologies Center, 3) the Biobank, and 4) the Data and Research Center. As of June 2015, the National Institutes of Health's All of Us website was updated to reflect that the Participant Technologies Center was split into two separate categories: Participant Centers and the Participant Technology Systems Center. As a result, All of Us is now divided into five areas.

For more information: www.allofus.nih.gov

Project Baseline

Project Baseline describes itself as "the quest to collect comprehensive health data and use it as a map and compass, pointing the way to disease prevention." Under the project, Verily has partnered with Duke University, Stanford Medicine, and Google to collect health data from 10,000 people for five years. Like the All of Us study, the Baseline study will be longitudinal and will collect multiple forms of health data including self-reported and measured data as well as sensor data. The participants in this study are engaged via a mobile app and an online interface. Anyone who is at least 18 years old, a resident of the United States, able to speak and read English or Spanish, and not severely allergic to nickel or metal jewelry may register online at project-baseline.com to enroll in the Project Baseline registry and be considered for the study. Verily is also an awardee for the Data and Research Center in the All of Us study.

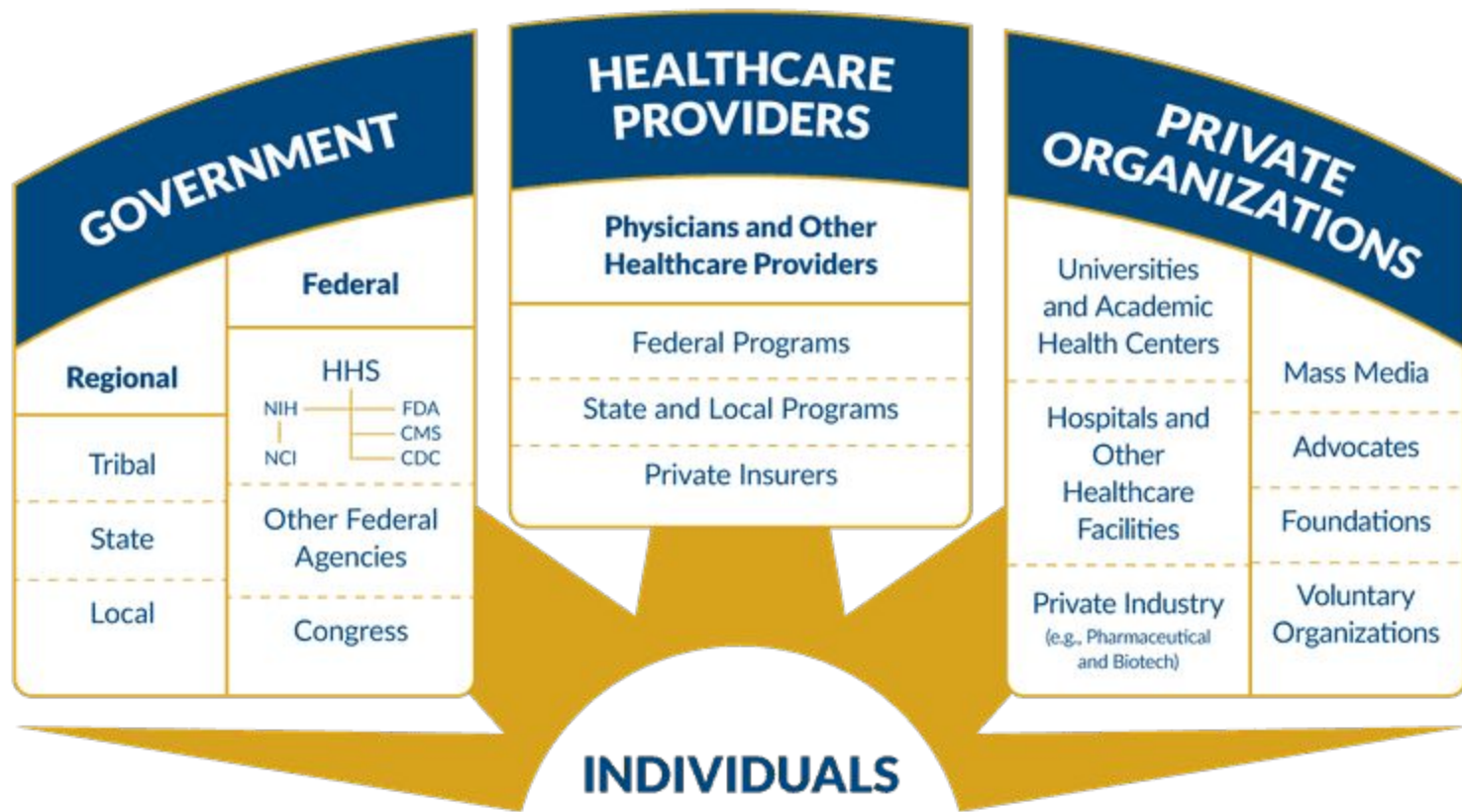
For more information: www.projectbaseline.com

American Heart Association My Research Legacy

The American Heart Association (AHA) is a non-profit organization that promotes cardiac care to reduce disability and deaths caused by heart-related problems. The AHA's Institute for Precision Cardiovascular Medicine is the only organization solely focused on precision medicine in cardiovascular care and sponsors the My Research Legacy Study. The study aims to collect health data from individuals who have had a cardiac event and focuses on heart conditions and stroke, unlike the All of Us Cohort Study and Project Baseline, which aim to gain insights about multiple diseases. The My Research Study is conducted in partnership with the Broad Institute of MIT and Harvard, an awardee in the All of Us study. Participant data will include data from online surveys and Fitbit Charge 2 devices and medical records if patients provide access. Some can also take part in a genetic sub-study that requires saliva samples. Only participants from Florida and New Jersey will get their genetic test results. The pilot study included 2,000 people and the main study will include 250,000.

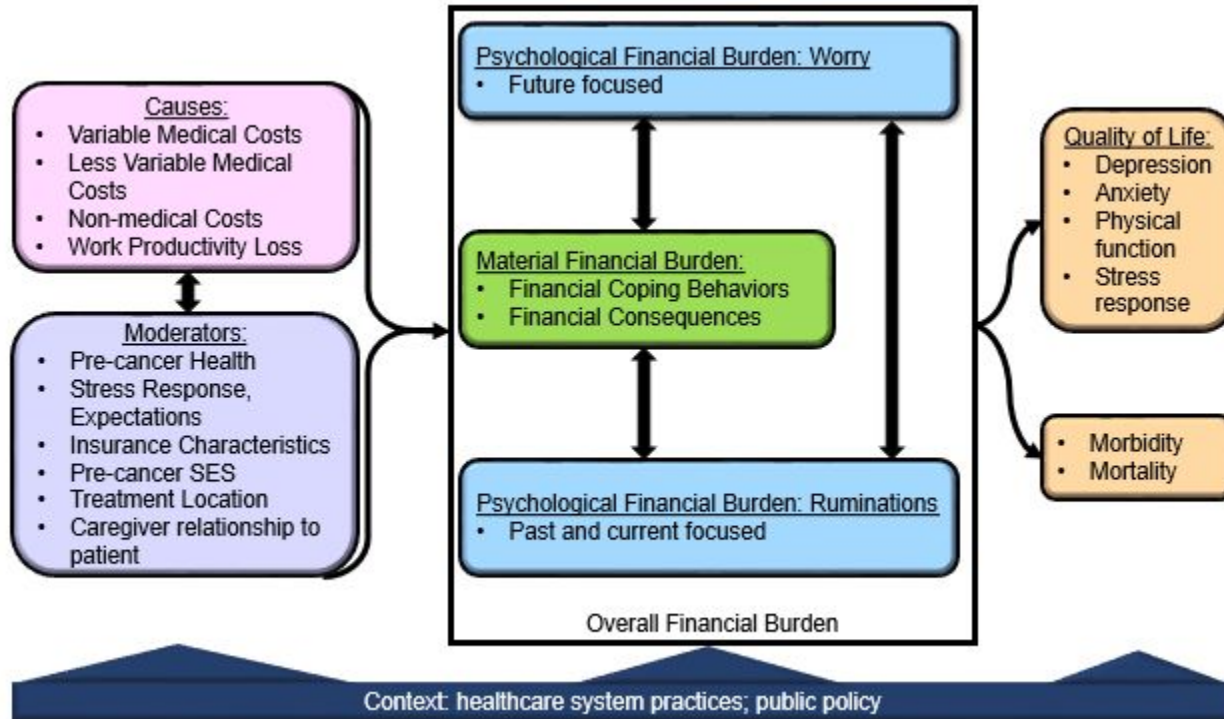
For more information: www.myresearchlegacy.org

President's Cancer Panel



Financial Burden

Theoretical Model of Financial Burden



The Financial Impact of Cancer

Did you know that financial distress from out-of-pocket cancer treatment costs increases the risk of death for individuals with cancer by 79%?¹

In 2022, 79% of our clients who reported their annual income made \$42,000 or less.

The monthly costs of basic needs for an average family in Texas:

- Rent = \$1,500
- Utilities & Phone = \$615
- Car Payment & Car Insurance for one car = \$630
- Gas for one car = \$250
- Groceries = \$800

How do you choose between groceries for your family or anti-nausea medication?

Anti-nausea medication and treatment equipment can cost \$200-\$750 per month, depending on the cancer type.

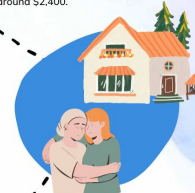


How do you choose between your car payment and cancer treatment?

Under an insurance plan, co-pays for radiation appointments can cost around \$60 each time – most radiation cycles are 40 sessions. The total cost of radiation treatment can total around \$2,400.

How do you choose between your basic needs and cancer?

We all can see how this amount of take-home pay will not cover basic monthly bills AND additional cancer-related bills.



Cancer Care Services exists, so no one has to choose between paying for basic needs and cancer-related bills.



PATIENT EDUCATION

Support Resources for Patients with Blood Cancers



Facing the realities of blood or bone marrow cancer can be difficult, but you don't have to go it alone. This guide has information on support groups and other helpful resources for patients of all ages and their families. Whether you want to connect with others who understand what you're going through, find other sources of emotional support, or access practical help, a broad range of resources is available. In addition, check with your local hospital to learn about groups or programs that may be open to you even if you receive your medical care elsewhere.

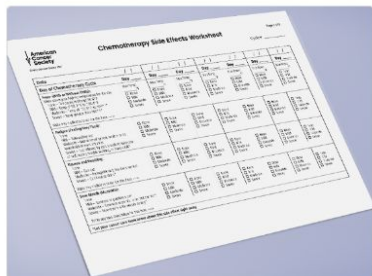
- * [Support for patients](#)
- * [General support](#)
- * [Language-specific resources](#)
- * [LGBTQ+ resources](#)
- * [Body image and beauty resources](#)
- * [Fertility resources](#)
- * [Camps and retreats](#)
- * [Caregiving services and support](#)
- * [Meals assistance](#)
- * [Transportation](#)
- * [Integrative medicine](#)

Cancer Care Toolkit

It can be hard to keep track of all the information you get about your cancer diagnosis and treatment. But keeping everything organized can help you feel less stressed and more prepared to talk with your doctor. We've created this printable cancer care toolkit to get you started.

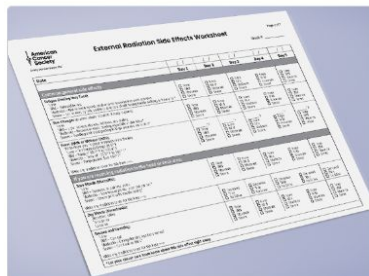
Worksheets and Trackers

Choose from this collection of worksheets to help you stay organized. Keep track of your appointments, medications, and test results all in one place. Print the pain diary and side effects worksheets to track your symptoms. Bring them along when you meet with your cancer care team.



Chemotherapy Side Effects Worksheet →

This worksheet covers 7 days of a chemotherapy cycle to track most common side effects.



Radiation Therapy Side Effects Worksheet →

This worksheet covers 1 week (5 treatment days) of radiation therapy to track most common side effects.



Pain Diary →

Pain may happen with cancer. It can affect many parts of your life. Using a diary with a pain scale helps you describe what you're going through.

My Lab Results

Use this form to keep track of your lab test results, as well as any notes about your labs.



cancer.org | 1.800.227.2345

Test	Reference or normal range	Date: XX/XX/XX	Date:	Date:	Date:	Notes/Questions
		Test result	Test result	Test result	Test result	
White blood cells (WBC)	5-10 x1000/mm ³	4.1 (L)				Infection-fighting cells
Hemoglobin (HGB)	12-18 g/dL	9				How much oxygen red blood cells carry
Platelets (PLT)	150-450 x1000/mm ³	100 (L)				Higher risk of bleeding when under 150

Every Bite Counts

Nutrition Tips for
Breast Cancer Survivors



GoFundMe

Stand with Liam — A Warrior's Second Battle with Leukemia



M Malea Estrada for Yvonne Levolt

🔒 Donation protected

Support Liam's Fight Against Leukemia (Round Two)

Liam Levolt is a brave and bright teenage boy from Costa Mesa, CA who loves playing sports, the trumpet, and making everyone around him smile. A few summers back on July 25, 2023, the Levolt family received the call every parent dreads, Liam's bloodwork showed dangerously high white blood cell counts, and his pediatrician told them to rush to Children's Hospital of Orange County (CHOC) for further testing.

Within hours, doctors diagnosed B-cell ALL Leukemia, and Liam began aggressive chemotherapy right away. After weeks in the ICU and months of difficult treatments, his cancer finally went into remission. The family was relieved to be moved to the oncology wing, surrounded by the incredible CHOC team who guided them through the first fight.



\$10,235 raised of **\$11K**

111 donations

Donate now

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M Marcos Sainz
\$485 · 2 mos

B Brian Roteliuk
\$200 · 10 d

L Lacey Miller
\$200 · 2 mos

S Sydney Ross
\$100 · 2 mos

M Maryam Rezai
\$100 · 2 mos

See all

☆ See top



Liam Levoit

Liam Levoit

14787 visits

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for Liam

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Levoit Family

Feb 15, 2026



Today's update

Happy Valentine's Weekend. We are now one day away from Liam's hospital admission. He will under go four days of radiation, 2 days of chemo, 1 day of rest. On Monday; February 23rd, Zara will be admitted to at 5 am to donate her bone marrow (approximately 1 liter), and Liam will receive the bone marrow transplant (BMT) later that afternoon. Zara will experience some discomfort but will be able to go home at the end of the day.

...

[Show more](#)

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♡ 8 17 comments ▾

Meal Train® for Levoit Family

 2 Adults, 2 Kids  Costa Mesa , CA  Drop-Off: Dinner 5pm



Thank you to our community for lifting us up throughout this unexpected challenge. We appreciate all of the meals and support!

We love the Levoit Family and want to support them and Liam as he fights ... [Read More](#)

Give a Gift Card

Recipient
Levoit Family



Organizer
Barb Inman



28 Participants
 A +22



Share

Recent Activity

 **Jeannie Kim**
Delivered a Meal 3 years ago



Experiment



Last Week

CLD 101

Causal Loop Diagrams (CLD) are great for:

- “Quickly capturing your hypotheses about the causes of dynamics.
- Eliciting and capturing the mental models of individuals and teams.
- Communicating the important feedback processes you believe are responsible for a problem.

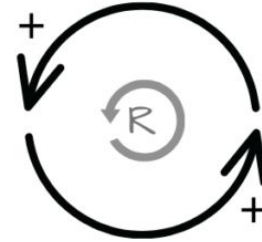


Figure 1.1 from [Müller, FMP, Springer 2015]

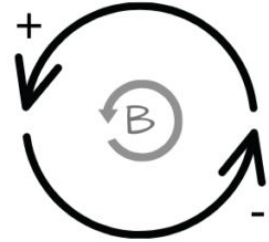
Feedback Loops

There are two kinds of feedback loops: reinforcing and balancing.

- In a **reinforcing loop**, the elements repeatedly perform the same action, resulting in exponential growth or decay.
- In a **balancing loop**, the elements of the system balance each other out.
- Both loops can bring along positive or negative consequences.



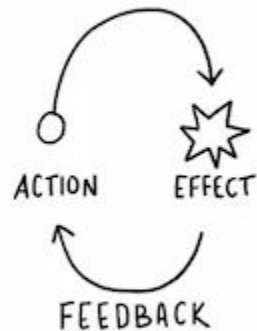
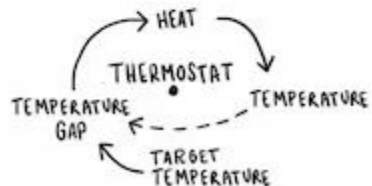
reinforcing
often results in
exponential increase/decrease



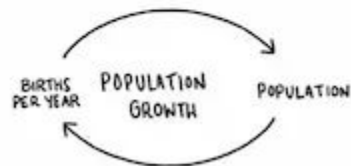
balancing
often results in
reaching a plateau

FEEDBACK LOOPS

- = +
BALANCING
PRODUCES STABILITY
SELF-CORRECTING



+ = -
REINFORCING
LEADS TO INSTABILITY
EXPONENTIAL GROWTH

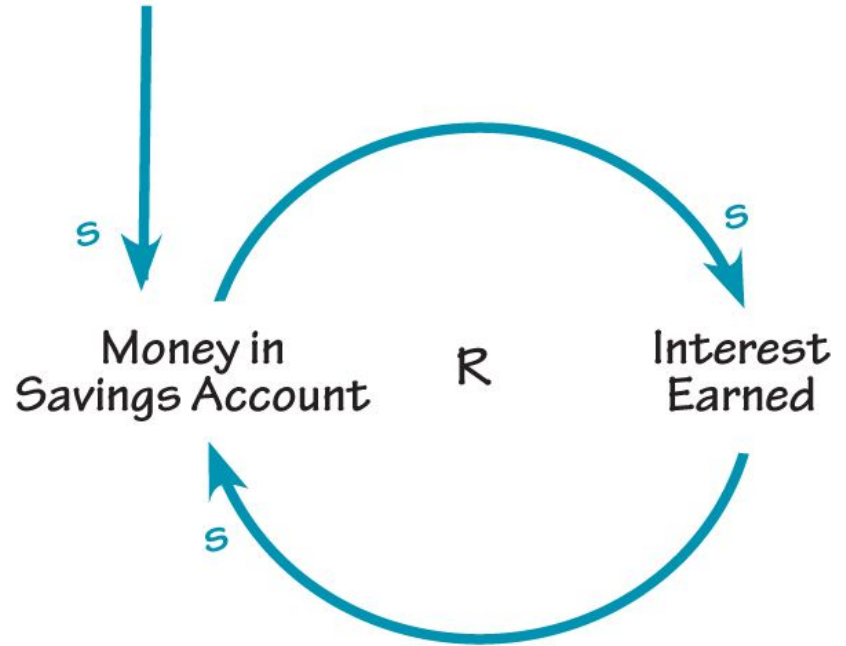


Cheat Sheet

- A **causal loop diagram** consists of four basic elements: the variables, the links between them, the signs on the links (which show how the variables are interconnected), and the sign of the loop (which shows what type of behavior the system will produce).
- The first step in creating a causal “story” is to identify the nouns or variables that are important to the issue. Remember, a variable is something that can vary over time.
- Draw an arrow from the cause to the effect.
- Indicate an increase (draw a "+") or a decrease (draw a "-") for the second element in each relationship.
- Create the loop diagram by adding all the variables and linking them with s (same direction) or o (opposite direction) connections.
- To determine if a loop is reinforcing or balancing, one quick method is to count the number of “o’s.” If there are an even number of “o’s” (or none are present), the loop is reinforcing. If there are an odd number of “o’s,” it is a balancing loop.
- When designing a causal loop diagram, we focus on the accurate illustration of an individual’s beliefs instead of capturing the world as it is.
- Systems mapping helps us gain new insights by organizing experiences and motivations in a relational framework.

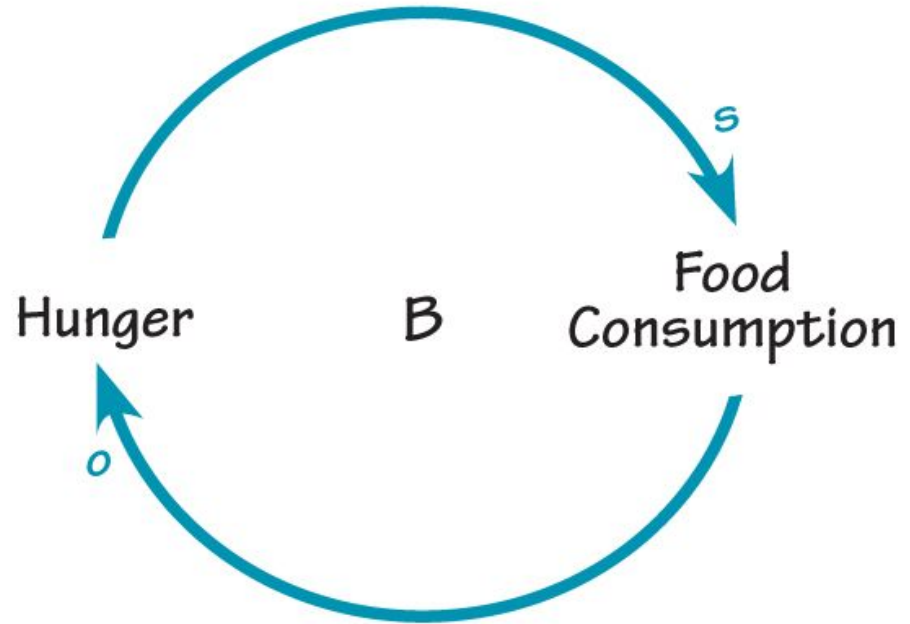
Example: Money

- **Reinforcing loops** change in one direction and is compounded by more change. S means the same direction connections.
- Draw an arrow from the cause to the effect.
- Money in a savings account generates interest, which increases the balance in the savings account and earns more interest.



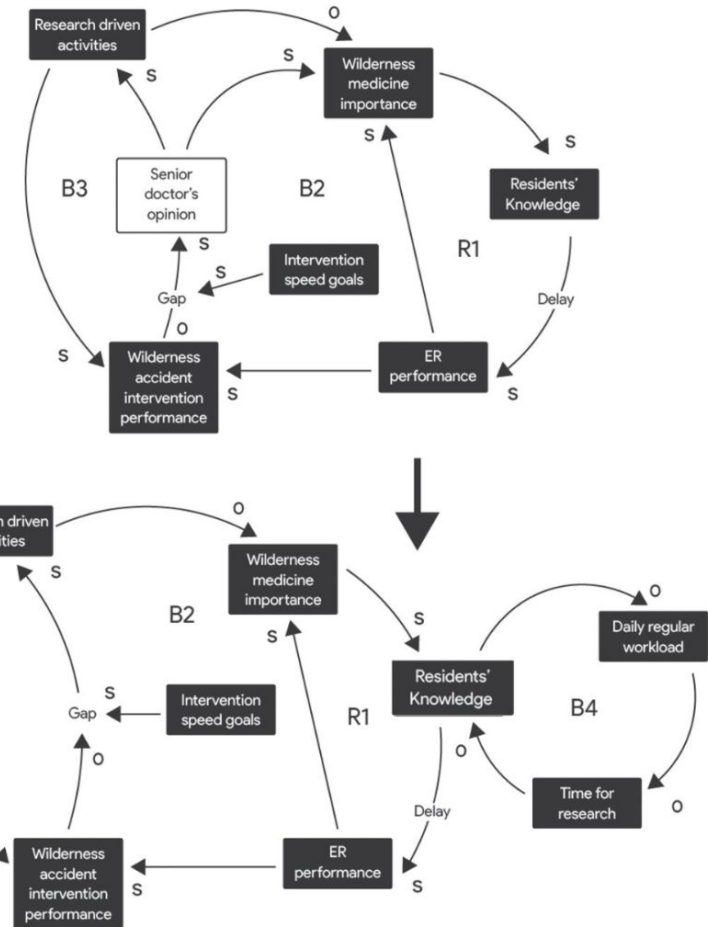
Example: Meals

- **Balancing loops** counter change in one direction with change in the opposite direction.
- Balancing processes attempt to bring things to a desired state and keep them there (thermostat).
- When we are hungry, our body sends a signal to our brain that it's time to eat, which appeases the hunger.



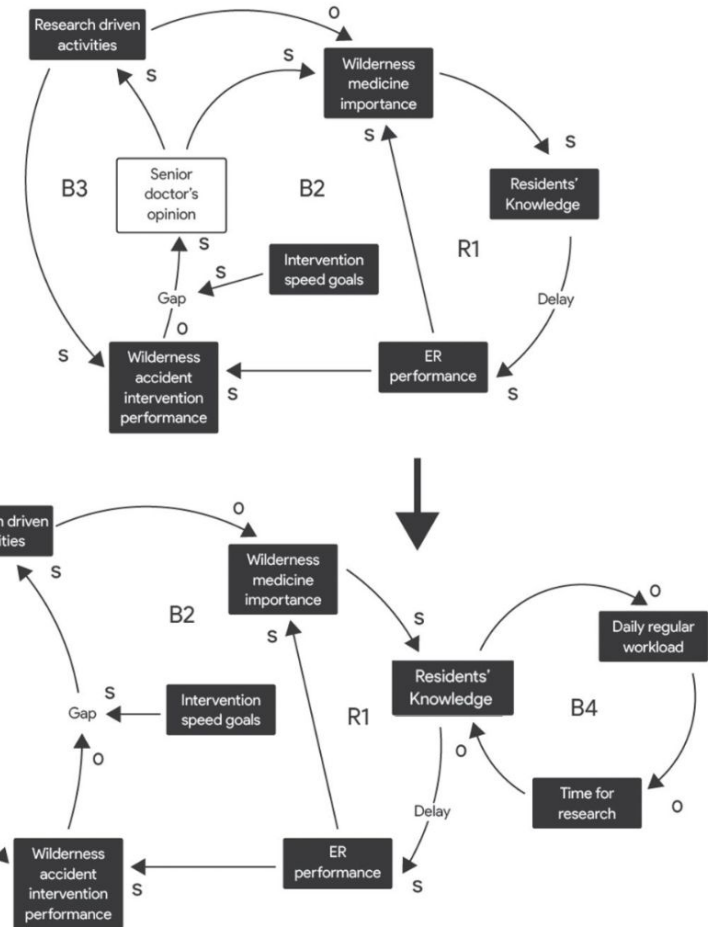
Example: Residents in Hospital

- Problem:** in one hospital, senior ER doctors asked a group of residents to improve their knowledge about wilderness medicine. The residents formed a wilderness medicine research team. The research team had to read through long documents, previous data on patient handling, and areas that lack proper resources.
- Key Elements:** Some of the most important elements could be the following: “Senior Doctors’ Expectations,” “Wilderness Medicine Improvement’s Importance,” or “Regular ER Duty Overload.”
- Relationship between Elements:** Look for causes, consequences, and constraints that bind these variables. We can connect “Too Much Work” with the “Lack of Time and Energy” variable with an S line. This means they change in the same direction, the more ER workload the residents have, the more they will lack the time and energy to do their research work.



Example: Residents in Hospital

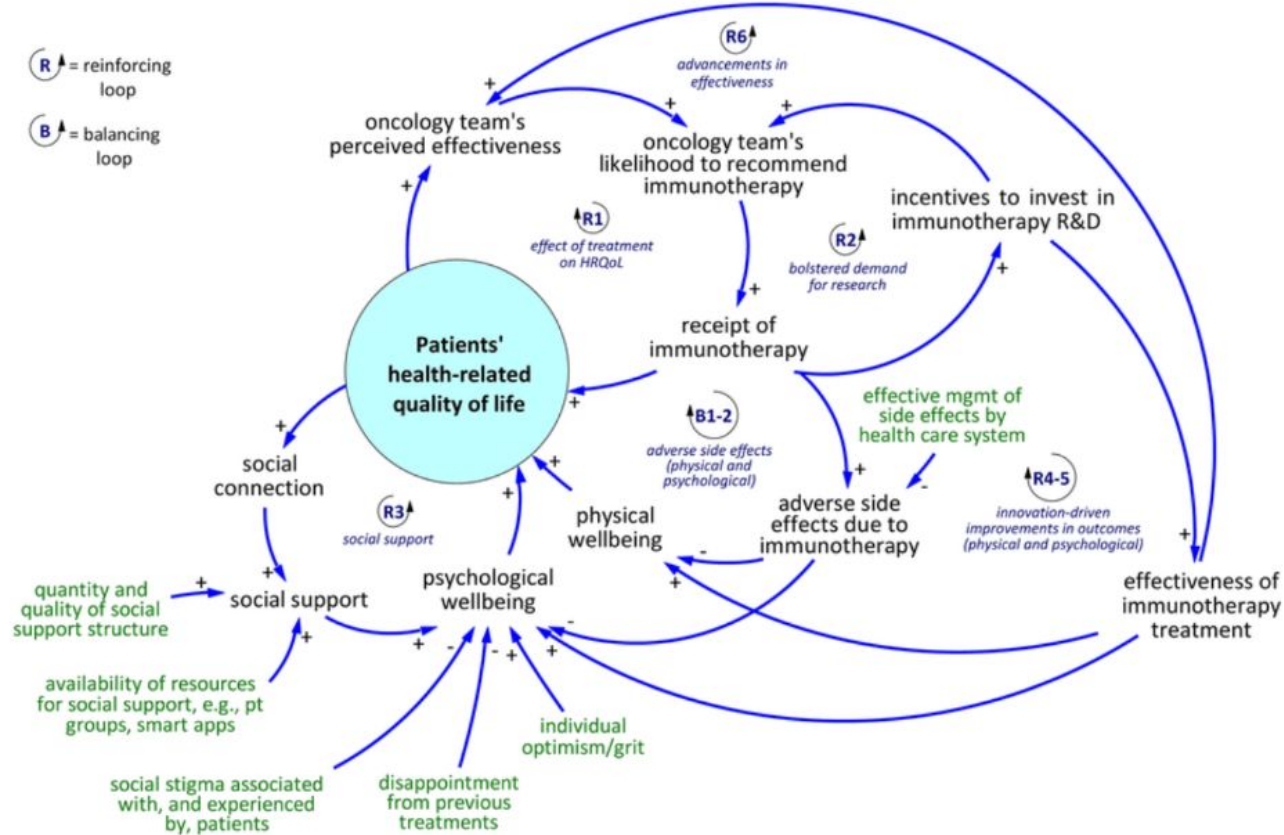
- **Tell the story:** The hospital story's title could be "Improvement without Time and Energy" to portray the resident team's dynamics. The research effort failed because the people who were responsible for it were overworked and tired. For the residents, the researching task was secondary to their more urgent daily demands.
- The first causal loop diagram (top) describes senior doctors' beliefs about the importance of wilderness medicine. Research and development about it would increase wilderness accident-related intervention quality and speed (R1).
- However, they were also focused on research-driven activities that would reduce the gap between wilderness accident intervention performance and the overall ER performance (B2 and B3).
- In the bottom diagram, Loop B4 is added to capture the dynamics of regular work demands that take the residents' time away from research activities. Senior doctors believed that as awareness and skillfulness of wilderness accident intervention increased, the ER doctors' quality of problem handling would also increase. This would eventually lead to improved long-term medical performance, further proving the importance of research and development in specific ER fields (Loop R1).



Instructions

- 1) **Choose the problem:** To get started with your causal loop diagram, you need to find an exact problem in a system you want to investigate. After finding the central problem, we need to collect information about its elements, interconnections, and boundaries.
- 2) **Select key elements and “scrub”:** Once you have the initial story of the problem, it's time to find the most critical elements. These could be identified easily if you prepared a cluster and interconnected circle map. Whichever elements ended up having the most links could be considered key elements.
- 3) **Describe the relationship between elements:** Step three requires you to find other variables that are closely connected to every significant variable. The more interconnections you identify, the better. Look for causes, consequences, and constraints that bind these variables.
- 4) **Tell the story:** We need to find several variable pairs to find a shared story running through them all. The shared story will guide us in developing a model that includes only the problem's relevant aspects. The story should have clear feedback mechanisms, which means the events presented should be recurring.
- 5) **Build a causal loop:** Create the loop diagram by adding all the variables and linking them with s (same direction) or o (opposite direction) connections. Add variables until the full story is told.
- 6) **Finalize the causal loop diagram:** When you're done sketching your diagram, double-check it to make sure it illustrates the story correctly. Track down logical fallacies if there are any. Make sure that all the cause-and-effect relationships are well presented.

Quality of Life in Cancer Immunotherapy



Thank you!
